

Highlights

OrbInspect: Coverage-Constrained Relative Trajectory Planning and ROS 2–Gazebo Validation for Autonomous Orbital Structure Inspection

- OrbInspect separates HCW dynamics authority from Gazebo visualization.
- A self-contained mesh benchmark reports trajectory, attitude, coverage, safety, and planner artifacts.
- The proposed set-cover CW tour reaches 98.33% mesh-target coverage.
- The best feasible method reduces Δv by 64.3% relative to the random-safe baseline.
- ROS 2 validation records synchronized control, coverage, planner, safety, and rosbag outputs.
- Gazebo Harmonic frames and videos validate visual replay, not orbital physics.

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Abstract

Autonomous inspection of large orbital structures requires a planner that can reason about relative orbital motion, sensor visibility, target coverage, collision avoidance, and fuel expenditure while remaining reproducible enough for system-level validation. This paper presents OrbInspect, a ROS 2 Jazzy simulation and evaluation framework for space-station-scale external inspection. OrbInspect uses a ROS-native Clohessy–Wiltshire/Hill dynamics node as the state source of truth, separates quantitative planning from Gazebo Harmonic visualization, and records trajectory, control, coverage, safety, planner, and configuration artifacts for paper-grade analysis. We formulate coverage-constrained inspection as a mixed visibility–dynamics problem and implement a mesh-aware offline planning experiment with area-weighted target sampling, line-of-sight filtering, attitude export, and seven planning methods. The best feasible method, a dynamics-aware weighted set-cover planner followed by a CW-transfer tour, reached 98.33% final coverage on 180 mesh targets with 357 candidate viewpoints, 23 selected viewpoints, 8.770 m/s cumulative Δv , 7.53 m minimum clearance, 0.564 m/s maximum speed, and 5.86 s planning time. The same benchmark showed that a pure coverage-greedy baseline also achieved 98.33% coverage but was infeasible because its trajectory entered the keep-out envelope, while a random-safe baseline required 24.57 m/s. A

ROS 2 execution smoke run produced synchronized trajectory, control, coverage, planner, safety, and rosbag artifacts, and Gazebo Harmonic validation using an International Space Station visual model confirmed replay, pose-following, and visual interface behavior without making Gazebo physics the authority for spacecraft dynamics.

Keywords: spacecraft inspection, relative orbital dynamics, coverage planning, next-best-view planning, Clohessy–Wiltshire equations, ROS 2, Gazebo Harmonic, reproducible robotics evaluation

1. Introduction

Large orbital structures are becoming central to civil, scientific, and commercial spaceflight. Space stations, assembled telescopes, logistics platforms, and in-space manufacturing facilities all require periodic external inspection to detect configuration changes, micrometeoroid damage, thermal blanket degradation, docking-interface anomalies, and other faults that are not easily observed from fixed onboard cameras. A small free-flying inspector can provide close-range, target-specific imagery, but autonomous inspection in the neighborhood of a large structure remains a coupled planning and validation problem: the vehicle must acquire useful views, satisfy relative orbital dynamics, remain outside keep-out regions, avoid excessive control effort, and produce evidence that the mission actually inspected the requested targets.

This paper presents OrbInspect, a ROS 2-based framework for orbital inspection planning, execution, and evaluation. The design is motivated by a practical research need. Inspection algorithms are often demonstrated either as stand-alone planners detached from middleware and visualization, or as

visually compelling simulators in which the quantitative coverage and safety evidence is difficult to audit. OrbInspect separates those roles. A ROS-native Clohessy–Wiltshire/Hill (CW/HCW) dynamics node is the source of truth for the chaser state; controllers, safety filters, coverage estimators, mission logic, and loggers are connected through ROS 2 interfaces; Gazebo Harmonic provides visual validation of the station and chaser scene but is not used as the orbital-dynamics authority.

The specific contribution of this manuscript is a coverage-constrained inspection-planning and validation pipeline. We formulate the inspection task in the local-vertical/local-horizontal frame, define target visibility and dwell criteria, impose actuation and keep-out constraints, and implement a mesh-aware offline planning experiment. The planner samples area-weighted mesh targets, generates candidate viewpoints, computes visibility with range, field-of-view, view-angle, and line-of-sight checks, scores viewpoints by new coverage and CW transfer cost, exports camera attitudes, and produces a bounded-acceleration CW reference trajectory. The resulting artifacts are stored in a self-contained experiment directory containing raw CSV files, configuration snapshots, figures, videos, and summary files.

The reported results show a substantial improvement over the earlier proxy planner baseline. On the mesh benchmark, the proposed set-cover CW tour achieves 98.33% coverage of 180 target samples with all transfers feasible, 8.770 m/s cumulative Δv , and 7.53 m minimum clearance. The comparison includes six baselines: proposed safe CW next-best-view, coverage-greedy, safe coverage-greedy, nearest next-best-view, fuel-greedy, and random-safe. A ROS 2 paper-experiment smoke run verifies the execution and logging

pipeline over trajectory, control, coverage, planner, safety, and rosbag topics. A Gazebo Harmonic validation path replays the saved offline trajectory and attitude stream with an ISS visual model, confirming simulator interfaces while leaving HCW dynamics as the state authority.

The main contributions are:

1. a ROS 2 Jazzy architecture for orbital inspection in which HCW dynamics remain the state source of truth and Gazebo is used for visualization;
2. a formal coverage-constrained relative trajectory planning problem combining visibility, dwell, fuel, speed, and clearance constraints;
3. a mesh-aware dynamics-aware set-cover CW tour planner with saved target, viewpoint, attitude, trajectory, coverage, and planner logs;
4. a self-contained validation package with figures, raw data, Gazebo frames, video artifacts, configuration snapshots, and summaries embedded under the paper directory;
5. an evidence-based comparison showing 98.33% coverage and lower Δv than random-safe and several greedy baselines.

2. Related Work

Orbital inspection draws on three bodies of literature: relative spacecraft motion, robotic view planning, and middleware-supported simulation. The Clohessy–Wiltshire equations remain a common starting point for rendezvous and proximity operations near a circular chief orbit because they provide a linear, interpretable model of relative translation [1]. Although high-fidelity perturbation models are required for flight design, a CW/HCW baseline is

appropriate for early algorithmic comparisons because it makes assumptions explicit and permits repeatable propagation, control, and transfer-cost calculations [2].

Robotic inspection and next-best-view planning study how a robot should select sensor poses that reconstruct or inspect an object efficiently. Classical surveys emphasize visibility, occlusion, sensor range, view quality, and the trade-off between coverage and motion cost [3]. Orbital inspection adds constraints that are less prominent in terrestrial settings: viewpoints must be dynamically reachable under relative orbital motion, safety margins must be maintained around delicate structure, and fuel consumption is a primary mission metric. This motivates planners that do not choose views solely by information gain, but also by transfer feasibility and clearance.

Simulation frameworks for space robotics must also be treated carefully. A rendered Gazebo scene can show useful visual context, but standard rigid-body physics in a robotics simulator is not a substitute for an orbital relative dynamics model. OrbInspect therefore uses Gazebo Harmonic for visualization of the ISS model and chaser while the ROS 2 dynamics node publishes the chaser state used by planning, control, safety, and logging. This separation follows the philosophy that numerical claims should come from logged planning and execution data, while visualization should support interpretation and interface validation.

The closest methodological position of this work is therefore not a single new optimizer in isolation, but an integrated, reproducible evaluation stack: coverage-aware viewpoint planning, CW-feasible trajectory generation, safety monitoring, ROS 2 execution, and Gazebo visualization are con-

nected through recorded artifacts that can be audited after each run.

3. System Model and Mission Setting

3.1. Architecture

OrbInspect is organized as a ROS 2 Jazzy workspace with packages for dynamics, control, guidance, perception, safety, mission management, evaluation, description, Gazebo visualization, and bringup. The system is designed around a strict authority boundary. The dynamics node propagates the chaser state under CW/HCW equations and publishes odometry. The controller tracks reference states and emits acceleration commands. The safety layer monitors clearance against station primitives and can filter commanded accelerations. The coverage layer evaluates which inspection targets are visible. The evaluation nodes record CSV logs, rosbag data, summaries, and figures. Gazebo Harmonic subscribes to the resulting state and renders the ISS model and chaser for visual validation.

3.2. Reference Frame and State

The inspection pass is represented in a local-vertical/local-horizontal frame attached to a nominal circular chief orbit. Over the time horizon considered here, the inspected structure is treated as fixed in this frame. The chaser translational state is

$$x = \begin{bmatrix} r_x & r_y & r_z & v_x & v_y & v_z \end{bmatrix}^T, \quad (1)$$

where $r \in \mathbb{R}^3$ is relative position and $v \in \mathbb{R}^3$ is relative velocity. The camera is assumed to point toward the inspected structure or toward a selected target

cluster during the baseline experiments; attitude slew constraints are left to later work.

3.3. Relative Dynamics

The baseline dynamics are

$$\dot{r}_x = v_x, \quad \dot{r}_y = v_y, \quad \dot{r}_z = v_z, \quad (2)$$

$$\dot{v}_x = 3n^2 r_x + 2nv_y + a_x, \quad \dot{v}_y = -2nv_x + a_y, \quad \dot{v}_z = -n^2 r_z + a_z, \quad (3)$$

where n is the chief mean motion and $u = [a_x, a_y, a_z]^\top$ is the commanded acceleration. The discrete propagation map is denoted

$$x_{k+1} = f_{\text{CW}}(x_k, u_k, \Delta t). \quad (4)$$

The same model is used in the offline trajectory generator and in the ROS dynamics node for the baseline validation reported here.

3.4. Inspection Targets

The station is represented for planning by a finite set $\mathcal{T} = \{1, \dots, N\}$ of target samples. Each target i has a point p_i , outward normal n_i , and accumulated dwell time τ_i . A target is considered inspected when it has remained visible for at least $\tau_{\min}$. The primary offline benchmark in this paper uses the NASA ISS mesh backend, area-weighted sampling with 180 target samples, and 357 candidate viewpoints. The ROS smoke run uses the online coverage configuration, which reports 142 target samples. Gazebo renders the ISS visual model for interpretation and trajectory replay; the quantitative coverage and dynamics claims are computed from the logged offline and ROS data.

3.5. Mission Termination

The mission starts from a safe initial relative state and terminates when a coverage threshold is reached, when the planner exhausts available useful viewpoints, or when a maximum mission time is reached:

$$C_K = \frac{1}{N} \sum_{i=1}^N y_{i,K}, \quad y_{i,K} = \mathbf{1}\{\tau_{i,K} \geq \tau_{\min}\}. \quad (5)$$

The high-coverage offline benchmark uses a 70% success threshold and a 98% planning stop ratio. The best feasible method reaches 98.33% final coverage, as reported in Section 7.

4. Problem Formulation

4.1. Visibility and Dwell

At time step k , target i is visible from chaser position r_k when range, field-of-view, incidence-angle, and line-of-sight predicates are satisfied:

$$\rho_{\min} \leq \|p_i - r_k\| \leq \rho_{\max}, \quad (6)$$

$$p_i \in \text{FOV}(r_k, q_k), \quad (7)$$

$$\arccos\left(\frac{(r_k - p_i)^\top n_i}{\|r_k - p_i\| \|n_i\|}\right) \leq \theta_{\max}, \quad (8)$$

$$\text{LOS}(r_k, p_i, \mathcal{O}) = 1, \quad (9)$$

where q_k denotes camera attitude or an equivalent boresight model, $\theta_{\max}$ is the maximum view angle, and $\mathcal{O}$ is the occluding geometry representation. The dwell update is

$$\tau_{i,k+1} = \begin{cases} \tau_{i,k} + \Delta t, & \text{if target } i \text{ is visible at } k, \\ \tau_{i,k}, & \text{otherwise.} \end{cases} \quad (10)$$

4.2. Safety and Actuation

The chaser must satisfy acceleration, speed, and clearance constraints:

$$\|u_k\|_2 \leq u_{\max}, \quad (11)$$

$$\|v_k\|_2 \leq v_{\max}, \quad (12)$$

$$d_{\mathcal{S}}(r_k) \geq d_{\text{safe}}, \quad (13)$$

where $d_{\mathcal{S}}(r_k)$ is the signed distance from the chaser to the station safety envelope. In the implemented ROS stack, a safety monitor logs minimum distance, clearance, caution-zone status, nearest primitive, filter activity, and nominal versus filtered acceleration commands.

4.3. Optimization Objective

The ideal finite-horizon inspection problem is

$$\min_{\{x_k, u_k, q_k\}_{k=0}^K} w_t K \Delta t + w_u \sum_{k=0}^{K-1} \|u_k\|_2 \Delta t + w_s \sum_{k=0}^K \phi(d_{\mathcal{S}}(r_k)) + w_c(1 - C_K) \quad (14)$$

$$\text{s.t. } x_{k+1} = f_{\text{CW}}(x_k, u_k, \Delta t), \quad (15)$$

$$\|u_k\|_2 \leq u_{\max}, \quad (16)$$

$$\|v_k\|_2 \leq v_{\max}, \quad (17)$$

$$d_{\mathcal{S}}(r_k) \geq d_{\text{safe}}, \quad (18)$$

$$y_{i,k} \text{ follows the visibility and dwell model}, \quad (19)$$

$$C_K \geq C_{\text{req}}, \quad (20)$$

where ϕ is a soft penalty used in relaxed formulations. This problem is mixed-integer and nonconvex because visibility, occlusion, dwell satisfaction,

and coverage are discrete or discontinuous. OrbInspect therefore implements a structured decomposition rather than a monolithic nonlinear mixed-integer program.

4.4. Evaluation Metrics

The primary metrics are final coverage, cumulative control effort, minimum clearance, maximum speed, and tracking error:

$$C_{\text{final}} = \frac{N_{\text{inspected}}}{N}, \quad (21)$$

$$\Delta v_{\text{cum}} = \sum_k \|u_k\|_2 \Delta t, \quad (22)$$

$$d_{\text{min}} = \min_k d_{\mathcal{S}}(r_k), \quad (23)$$

$$e_{\text{track,rms}} = \sqrt{\frac{1}{K} \sum_k \|r_k - r_k^{\text{ref}}\|_2^2}. \quad (24)$$

Additional reported metrics include planning time, selected-viewpoint count, candidate-viewpoint count, trajectory sample count, and safety-filter activation.

4.5. Dynamics-Aware Coverage Efficiency

The role of the CW model is not limited to post hoc feasibility checking. In the implemented offline planner, each candidate viewpoint is evaluated through a predicted bounded-acceleration CW transfer from the current state. This produces an edge cost

$$J_{\text{CW}}(x, z_j) = \lambda_v \widehat{\Delta v}(x, z_j) + \lambda_e \widehat{e}_{\text{track}}(x, z_j) + \lambda_s \max\{0, -\widehat{d}_{\text{min}}(x, z_j)\}, \quad (25)$$

where $\widehat{\Delta v}$ is the integrated acceleration norm, $\widehat{e}_{\text{track}}$ is the RMS transfer tracking error, and $\widehat{d}_{\text{min}}$ is the predicted minimum clearance. The planner

therefore prefers viewpoints that expose new surface area while requiring low control effort under orbital relative dynamics. The reported $\Delta v/C_{\text{final}}$ metric is used as a direct measure of coverage efficiency.

5. Offline Planner

5.1. Decomposition

The implemented planner decomposes inspection into mesh target sampling, candidate viewpoint construction, visibility matrix construction, viewpoint selection, CW tour generation, trajectory rollout, and camera attitude export. This decomposition is deliberately transparent: each stage writes artifacts that can be checked after the run. The self-contained benchmark includes method-comparison, viewpoint, planner, trajectory, attitude, and coverage CSV files.

5.2. Candidate Viewpoints

For each mesh target, candidate viewpoints are generated on offset shells and accepted only when they satisfy basic safety and view constraints. Each candidate z_j stores a safety margin and a precomputed set of visible targets. The resulting binary visibility relation can be written as

$$A_{ij} = 1 \iff \text{target } i \text{ is visible from candidate } z_j. \quad (26)$$

In the high-coverage offline benchmark, 357 candidates are generated for 180 mesh target samples.

5.3. Dynamics-Aware Viewpoint Selection

The proposed method first performs dynamics-aware weighted set cover and then orders the selected candidates with a greedy CW-transfer tour. At each set-cover iteration, the planner balances new surface area coverage against the CW transfer cost from the current predicted state. For candidate z_j , let ΔA_j be the newly visible mesh area and let $J_{\text{CW}}(x, z_j)$ be the transfer cost defined in Section 4. The set-cover stage uses the score

$$s_j = \frac{\Delta A_j}{1 + J_{\text{CW}}(x, z_j)} + w_m \min\{\hat{d}_{\min}(x, z_j), d_{\max}\}, \quad (27)$$

then updates the predicted state with the CW rollout to the selected viewpoint. A second ordering pass greedily chooses the next selected candidate with the lowest predicted CW dynamic cost from the current terminal state. This makes later choices depend on the orbital state reached by earlier choices, so CW dynamics affect both the selected viewpoints and their order. The logged `planner.csv` file records the selected candidate identifier, new target count, area gain, cumulative coverage, score, transfer Δv , dynamic transfer cost, minimum clearance, transfer tracking error, and feasibility flag for each selected viewpoint.

5.4. CW Transfer Generation

After a candidate is selected, the planner computes a bounded-acceleration CW transfer toward the viewpoint, samples the resulting trajectory, evaluates clearance and speed, and updates predicted coverage. The exported trajectory includes position, velocity, acceleration command, camera bore-sight, yaw, pitch, and quaternion samples, so the planned path specifies both where the chaser flies and how the inspection camera is pointed. Edges that

violate basic feasibility checks are penalized or rejected depending on the baseline method. The current implementation is still a tractable approximation rather than a global mixed-integer optimum; this limitation is discussed in Section 8.

5.5. Outputs

The planner outputs a timestamped reference trajectory

$$\mathcal{P} = \{(t_k, r_k^{\text{ref}}, v_k^{\text{ref}}, u_k^{\text{ref}})\}_{k=0}^K. \quad (28)$$

For the reported set-cover CW tour run, $\mathcal{P}$ contains 920 trajectory samples over 1840 s with an 80 s transfer duration and 2 s integration step. The trajectory and attitude stream are subsequently available to the ROS stack for replay, logging, RViz inspection, and Gazebo visualization.

6. Experimental Protocol

6.1. Offline Planning Benchmark

The primary quantitative planning artifact is the local high-coverage offline experiment directory. It uses a mesh geometry backend, CW/HCW dynamics, 180 area-weighted mesh targets, 357 candidate viewpoints, a 70% coverage success threshold, and a 98% planning stop ratio. The experiment compares `set_cover_cw_tour`, `proposed_safe_cw_nbv`, `coverage_greedy`, `safe_coverage_greedy`, `distance_greedy`, `fuel_greedy`, and `random_safe`. The planner writes raw CSV files, configuration snapshots, figures, and summary documents under the paper directory.

6.2. ROS 2 Execution Smoke Run

The ROS execution artifact is the local phase-10 smoke directory. The manifest records the `full_station` scenario, `greedy_nbv_safety_filter` method, and `hcw` dynamics backend. The run records odometry, nominal control, safe control, coverage map, mission event, safety status, and planner event topics. Its purpose is to validate that the ROS system produces the paper-grade artifact layout and that the safety and logging interfaces behave correctly. The updated validation launch can additionally replay the saved high-coverage trajectory and attitude stream for RViz and Gazebo visualization.

6.3. Gazebo Harmonic Validation

Gazebo Harmonic is used to visualize the station and chaser. The validation artifacts are stored in the local video-capture directory and include preview frames and videos from chaser-follow and full-ISS views. Gazebo does not determine the spacecraft state in these experiments. Instead, chaser pose is driven from the ROS state stream, preserving the principle that the dynamics node is the source of truth.

6.4. Monte Carlo Baseline Table

The primary method comparison is the deterministic mesh experiment described above. The older Monte Carlo directory is retained as a small regression artifact, but the paper’s quantitative claims are based on the high-coverage mesh comparison.

7. Results

7.1. Mesh-Aware Offline Planning

Table 1 summarizes the best feasible method in the high-coverage mesh experiment. The proposed set-cover CW tour reaches 98.33% coverage over 180 mesh targets, exceeding the 70% success threshold and nearly reaching the 98% stop target. The planner selects 23 viewpoints from 357 candidates and produces a 1840 s trajectory with 920 samples. Unlike the pure coverage-greedy baseline, every selected transfer is feasible and the minimum clearance remains positive.

Figure 1 overlays the proposed trajectory on the same ISS mesh model used for target sampling and Gazebo visualization. The figure is generated directly from the copied NASA ISS GLB, `raw/trajectory.csv`, and `raw/viewpoints.csv` in the saved high-coverage experiment, so the displayed path and selected viewpoints share the same LVLH coordinate frame as the mesh.

Figure 2 reports the corresponding coverage evolution for the deterministic mesh comparison. All curves are generated from the copied high-coverage result folder, keeping the manuscript figures tied to the archived CSV files.

7.2. Method Comparison

Table 2 reports the deterministic mesh comparison. All methods except the infeasible coverage-greedy baseline maintain positive clearance. The proposed set-cover CW tour has the lowest Δv among the methods that simultaneously achieve 98.33% coverage and remain feasible. It uses 64.3% less Δv

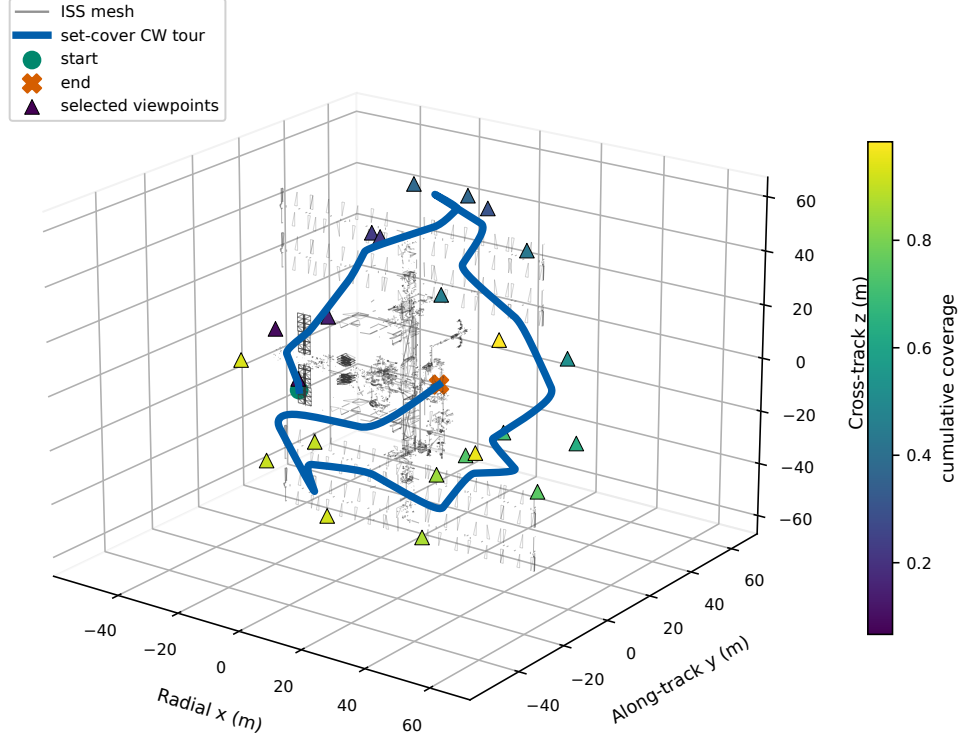


Figure 1: Proposed set-cover CW trajectory overlaid on the ISS mesh model. The selected viewpoints are colored by cumulative covered targets, and the thick blue curve shows the planned chaser path around the mesh.

than the random-safe baseline and 40.99% less Δv than safe coverage-greedy while retaining a much larger minimum clearance.

Figure 3 shows the saved 3D trajectories for all offline methods over the ISS mesh. The plot makes the dynamic consequence of each viewpoint ordering visible: methods with similar final coverage can still induce substantially different approach geometry, path length, and clearance behavior relative to the inspected structure.

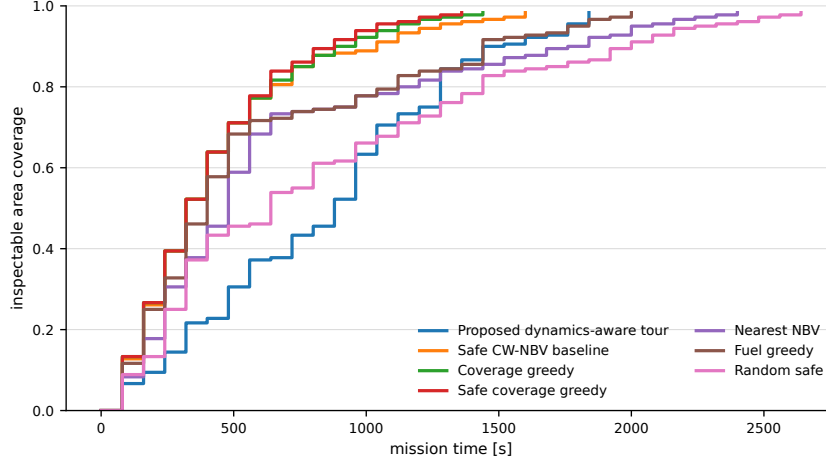


Figure 2: Coverage evolution for the compared offline methods in the high-coverage mesh experiment.

Figure 4 compares Δv , coverage efficiency, and minimum clearance. The set-cover CW tour is not the single best method for clearance, but it provides the best feasible trade-off among coverage, energy, and dynamic cost. The nearest-NBV baseline has the largest minimum clearance but requires more viewpoints, a longer mission duration, and higher Δv than the proposed method.

7.3. Trajectory, Attitude, and ROS Validation

The high-coverage experiment exports both trajectory and camera-attitude streams. For the set-cover CW tour, `trajectory.csv` and `attitude.csv` each contain 920 samples, while `viewpoints.csv` contains the selected camera viewpoints, boresight vectors, yaw, pitch, and quaternion orientations. These files are used by the trajectory replay node to publish odometry, path markers, field-of-view markers, and station mesh markers for RViz and

Gazebo validation.

The existing ROS execution smoke run remains a middleware validation artifact. It records trajectory, control, coverage, planner, safety, summary, manifest, and rosbag outputs. The run is shorter than the high-coverage offline trajectory, so its online coverage ratio is not used as the headline planning result. Its role is to demonstrate that the ROS stack records paper-grade topics and that Gazebo visualization consumes the ROS state stream while HCW dynamics remain the source of truth.

8. Discussion and Limitations

The updated results change the interpretation of OrbInspect from a preliminary proxy-geometry demonstration to a mesh-aware high-coverage planning baseline. The proposed set-cover CW tour reaches 98.33% coverage with positive clearance and the lowest Δv among feasible methods that achieve the same coverage. The comparison also clarifies why dynamics must be included in view planning: the pure coverage-greedy method achieves the same coverage but is infeasible because it violates the keep-out envelope. Coverage alone is therefore not an adequate objective for orbital inspection.

The proposed method is not universally optimal across every metric. The nearest-NBV baseline records a larger minimum clearance, and the proposed safe CW NBV method uses fewer selected viewpoints. However, both require more Δv than the set-cover CW tour. This indicates that the current weighted set-cover stage and CW tour ordering are effective for the chosen mesh benchmark, but the weight settings should be studied across additional geometries, initial states, target densities, and safety margins.

The mesh benchmark still uses a downsampled line-of-sight representation for occlusion checking. This is appropriate for a reproducible paper baseline, but a flight-oriented planner would need higher-fidelity geometry acceleration, uncertainty-inflated keep-out volumes, camera slew limits, illumination constraints, state-estimation error, plume constraints, and actuator saturation studies. The reported RMS transfer tracking error is also a planner-side rollout metric rather than a closed-loop flight validation error.

Gazebo validation should be interpreted conservatively. The videos and frames verify that the visual model, chaser model, camera view, and pose-following interfaces work in the integrated stack. They do not validate orbital mechanics, prove coverage, or certify collision avoidance. In OrbInspect, those quantitative claims come from the HCW dynamics, coverage, trajectory, attitude, safety, and planner logs.

8.1. Self-Contained Artifact Inventory

All cited figures and data are contained in the paper folder. The key local artifacts are listed in Table 3.

9. Conclusion

This paper presented OrbInspect, a ROS 2 and Gazebo Harmonic framework for coverage-constrained autonomous orbital inspection. The framework keeps CW/HCW dynamics as the state source of truth, uses Gazebo for visualization, and records the artifacts needed to audit planning, control, coverage, safety, and experiment configuration. The updated mesh-aware offline experiment generated a feasible 1840 s set-cover CW tour with 23 selected viewpoints, 8.7697 m/s planned cumulative Δv , 7.5323 m minimum

clearance, and 98.33% final coverage over 180 mesh targets. Compared with random-safe and several greedy baselines, the proposed method provides the best feasible coverage-energy trade-off in the reported benchmark. The ROS smoke run validated the logging and safety-monitoring pipeline, and Gazebo frames and videos validated the visual replay interface.

The next technical steps are broader Monte Carlo studies, higher-fidelity mesh occlusion acceleration, uncertainty-aware keep-out margins, camera slew and illumination constraints, and optional Basilisk validation after the HCW baseline remains reproducible. With those additions, OrbInspect can mature from a reproducible inspection-planning framework into a high-confidence experimental platform for autonomous inspection of large orbital structures.

References

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Algorithm 1 Dynamics-aware set-cover CW inspection planner

Require: Station mesh $\mathcal{S}$, initial state x_0 , camera model, d_{safe} , C_{req} , C_{stop}

Ensure: Planned trajectory $\mathcal{P}$ and inspection metrics

- 1: Sample area-weighted mesh targets $\mathcal{T} = \{(p_i, n_i, A_i)\}_{i=1}^N$
 - 2: Generate candidate viewpoints $\mathcal{V} = \{z_j\}_{j=1}^M$
 - 3: Reject unsafe candidates and compute visibility matrix $A \in \{0, 1\}^{N \times M}$
 - 4: Initialize covered set $\mathcal{C} \leftarrow \emptyset$, selected list $\mathcal{V}^* \leftarrow \emptyset$, and predicted state $x \leftarrow x_0$
 - 5: **while** $C(\mathcal{C}) < C_{\text{stop}}$ and the viewpoint budget is not exhausted **do**
 - 6: Roll out candidate CW transfers from x and compute ΔA_j and $J_{\text{CW}}(x, z_j)$
 - 7: Select $z^* = \arg \max_j \Delta A_j / (1 + J_{\text{CW}}(x, z_j)) + w_m d_{\text{min}}$
 - 8: Append z^* to $\mathcal{V}^*$
 - 9: Update x using the terminal state of the transfer
 - 10: Update covered set $\mathcal{C}$ from the candidate visibility vector
 - 11: **end while**
 - 12: Reorder $\mathcal{V}^*$ greedily using CW-transfer dynamic costs
 - 13: Simulate ordered trajectory and compute coverage, attitude, and safety
 - 14: Save trajectory, selected viewpoints, attitudes, visibility, and metrics
 - 15: **return** $\mathcal{P}$
-

Table 1: Best feasible offline planning metrics from
data/offline_high_coverage_experiment/summary.json.

Metric	Value
Method	set-cover CW tour
Dynamics model	CW/HCW
Geometry backend	mesh
Target count	180
Candidate viewpoints	357
Selected viewpoints	23
Trajectory samples	920
Mission duration	1840.0 s
Final coverage ratio	0.9833
Inspectable coverage ratio	0.9833
Coverage threshold	0.7000
Coverage success	true
Total planned Δv	8.7697 m/s
Δv per coverage	8.9183 m/s
Maximum speed	0.5641 m/s
Minimum clearance	7.5323 m
RMS transfer tracking error	28.6747 m
Planning time	5.8591 s
Feasible trajectory	true

Table 2: Offline mesh method comparison from `raw/method_comparison.csv`.

Method	Coverage	Δv	Clearance	Cost	Feasible
set-cover CW tour	0.983	8.770	7.532	152.8	true
proposed safe CW NBV	0.983	13.169	2.512	224.6	true
coverage greedy	0.983	15.665	-2.933	855.7	false
safe coverage greedy	0.983	14.865	1.307	251.0	true
nearest NBV	0.983	9.936	9.760	175.5	true
fuel greedy	0.983	11.362	2.582	199.4	true
random safe	0.983	24.565	2.123	409.1	true

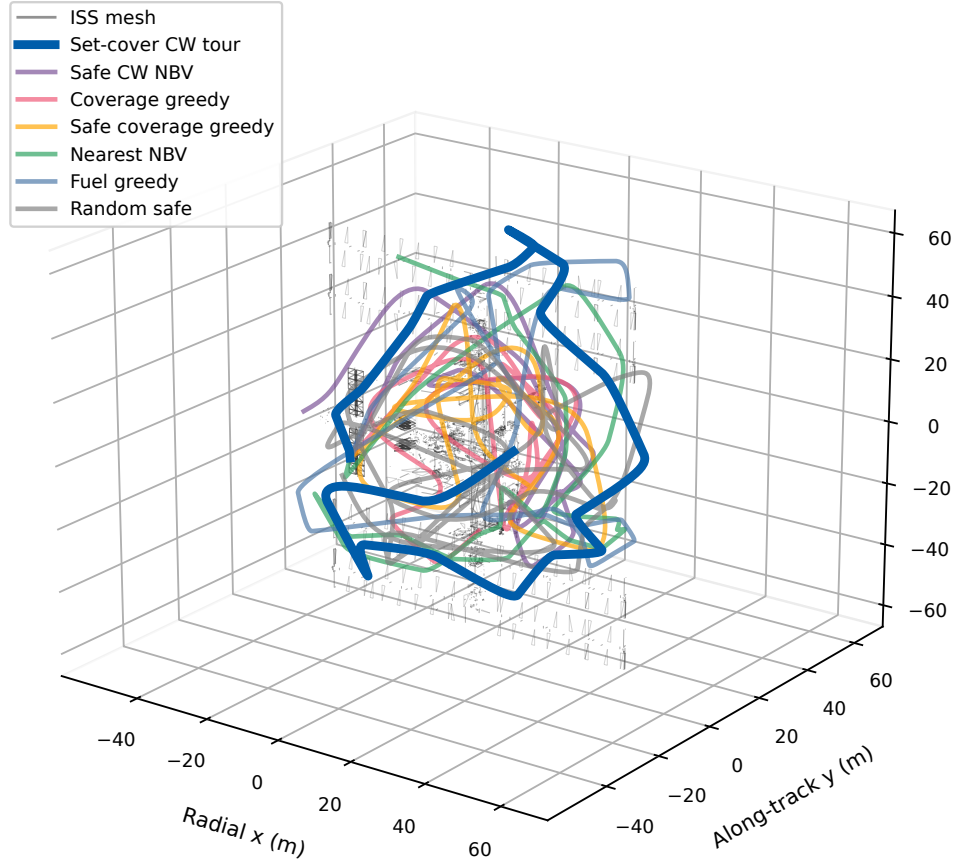


Figure 3: Offline trajectory comparison generated from `raw/trajectory.csv` and overlaid on the copied ISS GLB mesh. The proposed set-cover CW tour reaches the high-coverage target with the lowest Δv among feasible methods at the same final coverage.

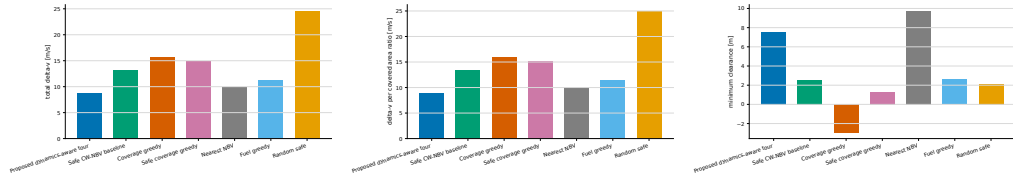


Figure 4: Offline comparison figures. Left: cumulative planned Δv . Center: coverage per unit Δv . Right: minimum clearance.



Figure 5: Gazebo Harmonic validation frames. Left: chaser-follow camera view. Right: full ISS visual context with chaser trajectory replay. Quantitative coverage, safety, and control claims come from CSV logs, not Gazebo physics.

Table 3: Self-contained manuscript artifacts.

Local artifact	Contents
High-coverage figures	Mesh-overlaid trajectory, coverage, Δv , efficiency, and safety comparison figures.
ISS mesh figures	High-resolution NASA ISS mesh preview in PDF, PNG, and SVG form.
ISS mesh model	Copied NASA ISS GLB used to regenerate the mesh-overlaid trajectory figures.
High-coverage data	Summary files, config snapshot, and raw comparison, planner, viewpoint, attitude, trajectory, and coverage CSV files.
Gazebo figures	Gazebo preview frames and ROS smoke-run trajectory/control figures.
ROS smoke data	Summary files, experiment manifest, config snapshots, and raw ROS execution CSV files.
Legacy offline data	Earlier proxy and mesh single-run planner artifacts retained for traceability.
Video data	Gazebo validation videos and preview frames.